BMS Information & Interaction Design Principles

A Comprehensive Blueprint for Business-First UI & HCI Excellence

Part 1 — What to Show and How to Show It

These core design principles dictate product behavior, information hierarchy, and task flow across the Business Management System (BMS). Every screen must directly serve operational efficiency and align with modern retail and SME practices.

1. Design for the business, not the software

Every screen should read like it was built by someone who runs a shop, not someone who writes code. If a field, label, or concept only makes sense to an engineer, it doesn't belong in the UI. Group screens around business questions ("How did today go?", "What do I need to reorder?"), not around database tables (Products, Transactions, Ledger Entries).

2. Show only what the user needs, when they need it

Every extra field or number on screen is a small tax on attention. Default to less, and let people opt into more. A dashboard alert like "3 products low on stock" is enough on the main screen; the full list lives one click away, not inline.

3. Speak the language of business owners, not developers

No technical vocabulary leaks into the interface — not in labels, not in error messages, not in empty states. "We couldn't save this — check your connection and try again," not "Sync failed: 409 conflict."

4. Every screen should help the user complete meaningful work

A screen earns its place only if it moves someone toward finishing a real task — closing a sale, restocking, reconciling the day. A low-stock list isn't just a list — it should let the owner reorder or message a supplier right from that screen. Screens that only display data without enabling a next action are usually a sign the task is missing, not the data.

5. Consistency builds confidence

Once a pattern is learned in one part of BMS, it should work identically everywhere else — same icon meaning, same button placement, same wording for the same action. "Void" always means the same thing and looks the same way, whether it's voiding a sale, an invoice, or a stock adjustment.

6. The system should adapt to the business, not the other way around

Kenyan SMEs vary — a supermarket, a pharmacy, a hardware shop. The interface shouldn't force one workflow onto all of them. Only show tax fields if the business has registered for VAT; let an owner reorder or hide dashboard cards that don't matter to their business type.

7. Numbers earn their place by relevance, not by availability

Just because a metric can be calculated doesn't mean it deserves screen space. Ask "would an owner change a decision based on this?" before adding it.

8. Default to now, let people ask for history

Show today by default — today's sales, today's cash position, today's low stock. Historical views and comparisons are one deliberate action away, not the default view competing for attention.

9. One primary next step per screen

Even data-dense screens should have a single, obvious "what to do here" action. If a screen has three equally-weighted CTAs, that's a sign the screen is doing two jobs and should split.

10. Errors and empty states teach, they don't just report

An empty inventory list should explain how to add the first product, not just say "No products found." A validation error should say what to type instead, not just that the input was wrong.

Part 2 — Foundational HCI Principles

Speed & Interaction Efficiency

- **Fitts's Law:** The time to hit a target depends on its size and distance from the current pointer/finger position. Frequently-tapped touch targets (Complete Sale, payment method buttons) need generous hit areas positioned where the hand naturally rests after the prior action, not wherever they fit in a generic layout.

- **Hick's Law:** Decision time increases with the number of choices presented. A quick-action grid with dozens of equally-weighted buttons is slower to scan than one with a handful of curated favorites plus a search field.
- **Doherty Threshold:** If the system responds in under ~400ms, the user stays in flow; above that, they mentally disengage and start double-checking. Scanning a barcode or searching a product needs to feel instant, or perceived responsiveness (optimistic UI, skeleton states) has to fill the gap.

Cognition & Perception

- **Miller's Law (chunking):** Working memory holds about 7±2 items. Long forms should be chunked into visually distinct groups of 3–5 related fields rather than one long scroll.
- **Recognition over recall (Nielsen):** People find it far easier to recognize something on screen than to remember it from memory. Favors autocomplete over free-typed codes, visible recent-transaction lists over asking a cashier to remember a receipt number, and a command palette that surfaces likely actions rather than requiring exact command names.
- **Gestalt grouping (proximity & common region):** Items placed close together or inside a shared border are perceived as related, even without a label. Used deliberately, not accidentally, to organize dense dashboards and cart layouts.
- **Von Restorff effect (isolation):** An item that visually deviates from its surroundings is noticed and remembered disproportionately. The one thing that truly needs attention (a stockout alert, an overdue reconciliation) should actually look different — not just differently colored within an otherwise identical card grid. Corollary: don't let a lower-risk action accidentally out-pop a higher-risk one through color alone.

Trust, Error Handling & Recovery

- **Error prevention over error messages (Nielsen):** Better than a good error message is making the error impossible. A numeric keypad for quantity entry prevents the error rather than catching it after the fact.
- **User control & freedom (undo over confirm):** Confirmation dialogs suit high-consequence, hard-to-reverse actions (void sale, delete product, restore backup). For medium-risk, reversible actions, an "Undo" toast after the action is often faster and less friction-heavy than a dialog before it — decide per action which pattern fits, rather than defaulting to one everywhere.

System Model

- **Tesler's Law (conservation of complexity):** Every system has an irreducible amount of complexity; the only question is who absorbs it. Push that complexity onto the owner, once, during setup, rather than letting it leak into the cashier's daily, repeated screen. This is the mechanism behind Principle 6 — it explains why configuration belongs in Settings and never in the POS workspace.

- **Match between system and mental model:** Navigation and data structure (organization → business → branch) should mirror how an owner actually thinks about their business, not how the database happens to be normalized.